

Educational Qualifications & Academic Achievements			
Course	Institution	Year	Grade
B.A. (Hons.) Economics	Sri Guru Tegh Bahadur Khalsa College (Delhi University)	2024-27	8.27 CGPA (1 st Year)
Class XII	St. Mary's Convent Sr. Sec. School (CBSE)	2023-24	96.6%
Class X	St. Mary's Convent Sr. Sec. School (CBSE)	2021-22	88.8%
- Awarded School Topper and subject excellence trophies for Mathematics, Economics, and Business Studies (Class 12). - IIM Bangalore – Completed Year 1 of B.B.A (Digital Business & Entrepreneurship). - Secured 99th Percentile in Accountancy (CUET – 2024).			

Independent Research & Working Papers
Causal Returns to Education in the Informal Economy: An Instrumental Variables Approach
Interactive Replication Report: https://mohanbhavya98.github.io/plfs-returns-to-education/ - Cleaned large-scale PLFS 2023–24 microdata to construct robust wage, education, and demographic control variables for empirical analysis. - Addressed endogeneity in human capital by estimating causal wage effects via an Instrumental Variables (IV) approach utilizing household data. - Programmed high-dimensional fixed-effect models and interaction terms using fixest and AER in R to analyze heterogeneous wage returns.
Labour Market Segmentation and Wage Penalties: A Baseline Empirical Analysis (SSRN Working Paper)
SSRN Preprint: http://ssrn.com/abstract=6635379 - Analyzed wage determinants and formal-informal sector differentials using unit-level microdata from the Periodic Labour Force Survey (PLFS) 2019–20. - Estimated baseline Mincerian OLS earnings functions in R to Identify structural human capital wage penalties across segmented labor markets.
An Empirical Re-examination of the Expectations-Augmented Phillips Curve (SSRN Working Paper)
SSRN Preprint: https://ssrn.com/abstract=6635519 - Conducted a rigorous structural break analysis utilizing quarterly US macroeconomic time-series data extracted from FRED (Q1 2009–Q2 2025). - Evaluated Phillips Curve non-linearity across distinct macroeconomic regimes to empirically test inflation-unemployment trade-off convexity.

PROJECTS
Dynamic Asset Allocation via Macroeconomic Regime Switching (Quantitative ML Project)
GitHub Repository: https://github.com/mohanbhavya98/macro_regime_hmm_research - Engineered an unsupervised Gaussian Hidden Markov Model (HMM) in Python to mathematically Identify macroeconomic regimes from time-series data. - Applied machine learning algorithms to optimize dynamic asset allocation strategies based on probabilistic state transitions.
Hikikonect – Agile Coil Program (Hiroshima University, Japan) Feb 2025 – Mar 2025 (2 months)
<i>Developed "Hikikonect," a social entrepreneurship project under the Agile Coil Program to help youth combat social isolation in Japan.</i> - Collaborated with students from Japan and the UK to develop Hikikonect, a social initiative addressing youth social isolation (“Hikikomori”) in Japan. - Conducted market and social research to identify key causes of isolation and designed community-based engagement solutions. - Created a digital outreach model integrating awareness campaigns and peer support tools to enhance social inclusion - Shortlisted for the in-person workshop at Hiroshima University for demonstrating creativity, collaboration, and social impact.
Economic Data Analysis Project – Delhi University (SEC Data Analysis) Mar 2025 – Apr 2025 (1 months)
<i>Analyzed employee-level data from a private firm to study compensation and workforce patterns using Excel and Python.</i> - Utilized Excel and Python to perform exploratory data analysis (EDA) on workforce data involving <i>CTC, experience, gender, sector, and roles</i>. - Conducted data cleaning, correlation, and regression analysis to evaluate relationships between compensation and work experience. - Designed interactive visual charts and dashboards in Excel to highlight wage and representation disparities across departments.

LEADERSHIP & POSITIONS OF RESPONSIBILITY
Learning Pod Leader – IIM Bangalore (BBA Digital Business & Entrepreneurship) Dec 2024 – July 2025 (8 months)
- Selected as 1 of 60–70 pod leaders from a batch of 1,000+ students for outstanding academic performance in key subjects. - Facilitated interactive academic discussions to strengthen conceptual clarity and peer learning. - Organized and led virtual learning meets, promoting academic collaboration and improving subject comprehension

TECHNICAL PROFICIENCY & CERTIFICATIONS
Technical Skills
Programming & Tools: R (Advanced data wrangling, causal inference), Python (pandas, NumPy), LaTeX, Quarto, GitHub, Advanced MS Excel. Econometrics & Statistics: Instrumental Variables (IV), Ordinary Least Squares (OLS), High-Dimensional Fixed Effects, Hypothesis Testing. R Packages: tidyverse, dplyr, fixest, AER, modelsummary, ggplot2. Research Methods: Large-scale microdata cleaning (e.g., PLFS), literature reviews, baseline empirical modeling.
Certifications
Data Analytics and Visualization (Accenture – Forage) Excel for Business (Goldman Sachs – Forage)

COLLEGE SOCIETIES & ECA	
Khalsa Street (Finance & Investment Cell)	- Helped in making industry reports & one-pager report. - Explored key financial concepts and contributed to finance newsletters as part of the Research & Development department. - Formatted company Financial Statements & analyzed using Ratio Analysis
Competitions	Secured Rank 2 in the Accounting Olympiad organized by Christ University, Delhi NCR, demonstrating strong conceptual understanding of accounting principles. - Achieved School Rank 1 and International Rank 497 in the International Accounting Olympiad conducted by Science Olympiad Foundation (SOF). - Received a Certificate of Appreciation signed by Hon’ble PM Narendra Modi for essay writing in Pariksha Pe Charcha 2021.